

Kathmandu University School of
Management (KUSOM)
Bachelor of Business Administration (BBA)



Curriculum of
MAS: 105 Business Mathematics II

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CHAPTER 1

COURSE INTRODUCTION

Kathmandu University School of Management (KUSOM) introduced its Bachelor of Business Administration (BBA) program at the central campus in Dhulikhel with a vision to build foundational corporate excellence. As part of this academic framework, the Department of Management Informatics and Communication (DoMIC) implements the **Business Mathematics II** curriculum to enrich undergraduate personal and professional development. This quantitative course is specifically designed to sharpen analytical thinking, allowing future managers to systematically address corporate challenges and execute data-driven decisions. To maintain educational relevance, DoMIC continuously re-engineers its academic frameworks to match evolving corporate expectations. Designed under the overarching educational policy of Kathmandu University (KU), this document presents a revised, comprehensive seven-module curriculum roadmap explicitly structured for first-year, second-semester BBA students.

1.1 Course Profile

This section outlines the basic profile of the course. Students learn advanced mathematical tools focusing on calculus and its applications under this course plan. It is a **3-credit hour** course, engaging facilitators and learners for **48 hours** altogether within the semester.

- **Course Title:** Business Mathematics II
- **Course Code:** MAS 105
- **Credit Hours:** 3 Credits
- **Total Contact Hours:** 48 Hours (*excluding assessment and final examination*)

1.2 Objectives

Business Mathematics II is the application of continuous mathematical tools and techniques to solve complex business, information systems, and managerial problems. It integrates concepts from differential, integral, and multivariable calculus to support structured decision-making in areas such as finance, optimization, resource allocation, and operations management. In modern data-driven business environments, calculus plays an essential role in algorithmic modeling, business intelligence, and economic optimization.

Main Objectives:

The objective of the course is to give a sound basic knowledge of calculus that is applicable in Economics, Quantitative Techniques, Finance, and other related fields. The specific learning objectives to be achieved are:

1. To provide a rigorous understanding of continuous functions, limits, and rates of change.
2. To introduce foundational concepts of differential calculus for single and multi-variable equations.
3. To develop applications of derivatives in optimization and elasticity problems within economics.
4. To familiarize students with integral calculus techniques for accumulated values and area optimizations.
5. To enable students to set up, solve, and interpret first-order differential equations.
6. To equip students with practical skills in multi-variable business optimization problems using Lagrange multipliers.

1.3 Course Engagement

The volume of learning and facilitating for this course is 48 hours. It is primarily run through face-to-face contact sessions and backed by offline collaborative reading practices.

- **Face-to-Face:** Main interactive contact hours consisting of concept lectures, immediate problem solving, and group participation.

- **Offline:** 100% of the parallel weight is expected for independent learning, assignments, working through textbook problem sets, and preparation for presentations.
- **Technology Integration:** Graphing calculators or any relevant computer software (such as Python or spreadsheet tools) will be actively utilized in the classroom to ensure students become well-versed in generating and interpreting mathematical output.

1.4 Prerequisites

The core prerequisites for learners taking this course include basic corporate literacy, a foundational command of algebra, systems of linear equations, functions, graphs, and the non-calculus concepts covered in **Business Mathematics I**.

1.5 Learning Outcomes (MLOs)

Upon successful completion of this course, students will be able to:

- **MLO 1:** Evaluate limits algebraically and graphically, and pinpoint points of continuous or step-wise business discontinuities.
- **MLO 2:** Apply foundational rules of differentiation to find rates of change for marginal analysis of revenue, cost, and profit functions.
- **MLO 3:** Calculate price elasticity of demand and interpret its revenue implications.
- **MLO 4:** Utilize first and second derivatives to sketch economic curves and isolate absolute optimization limits.
- **MLO 5:** Use integration techniques to solve initial condition problems and compute market efficiencies (Consumer and Producer Surplus).
- **MLO 6:** Construct and solve basic first-order differential equations representing economic growth or asset depreciation.
- **MLO 7:** Work with functions of several variables, computing partial derivatives and applying Lagrange multipliers to solve constrained management problems.

CHAPTER 2

COURSE PLAN

2.1 Course Outline

The structural progression of the 48-hour timeline across the six distinct learning modules is mapped below:

Units	Hours	Core Sub-Units & Topics
Unit One: Limits and Continuity	5 Hours	Properties of Limits, Limits at infinity, Continuity, Locating Discontinuities in Business Functions.
Unit Two: Differentiation	10 Hours	The Derivative and Derivative as a Rate of Change, Rules for Differentiation (Product, Quotient, Chain, and Power Rules), Differentiability and Continuity, Derivatives of Logarithmic and Exponential Functions, Elasticity of Demand, Implicit and Logarithmic Differentiation, Higher-Order Derivatives, Finding Root of Equation using Newton's Method.
Unit Three: Curve Sketching & Applied Maxima/Minima	8 Hours	Relative Extrema & First Derivative Test, Curve sketching using first derivative test, Absolute Extrema on a Closed Interval, Concavity and inflection points, Second Derivative Test, Asymptotes, Applied Maxima and Minima (Cost/Profit Optimization).

Unit Four: Integration	9 Hours	Indefinite Integral & Indefinite Integration with Initial Conditions, More Integration Formulas & Techniques of Integration, Summation notation properties, Definite Integral & Fundamental Theorem of Integral Calculus, Area between curves, Consumer's and Producer's Surplus
Unit Five: Methods & Applications of Integration	6 Hours	Integration by Parts, Integration by Partial Fraction, Introduction to Differential Equations, Applications of Differential Equations (Growth/Decay Models)
Unit Six: Multivariable Calculus	10 Hours	Functions of Several Variables, Partial Derivatives & Higher-order Partial Derivatives, Application of Partial Derivatives & Implicit Partial Differentiation, Multivariable Chain Rule, Maxima and Minima for Functions of Two Variables, Lagrange Multipliers , Multiple Integrals

2.2 Lesson Plan

Lecture (2 Hrs.)	Topic Covered	Objectives
Unit 1: Limits and Continuity (5 Hours)		
L1	Introduction to Limits & Evaluation • Intuitive and graphical idea of limits • Algebraic properties of limits • Evaluating indeterminate forms	Understand functional behavior near a point graphically and numerically, evaluate limit of polynomial and rational functions.
L2	Limits at Infinity & Functional Continuity • Limits involving infinity ($x \rightarrow \pm\infty$) • Mathematical definition of continuity • Testing continuity at a point and over intervals	Evaluate limit behavior at infinity, and test the three formal conditions of continuity.
L3 (Part 1)	Locating Discontinuities in Business • Types of mathematical discontinuities • Application to step-rate tax and tiered pricing	Identify types of mathematical breaks such as jump, removable, and infinite, interpret their specific business meanings.
Unit 2: Differentiation (10 Hours)		
L3 (Part 2)	The Derivative Concept • Derivative as a limit and geometric slope • Instantaneous rate of change concept	Interpret the derivative as a tangent line slope, understand its function as a marginal economic metric.
L4	Basic Differentiation Rules & Core Techniques • Constant, sum, and power rules • Product rule and quotient rule • Differentiability versus continuity	Differentiate standard algebraic equations using fundamental linearity rules, compute derivatives of products and quotients of expressions.

L5	Composite & Transcendental Functions • The Chain Rule and extended power rule • Derivatives of logarithmic functions • Derivatives of exponential functions	Differentiate nested composite functions, find rates of change for exponential growth and continuous compound interest equations.
L6	Advanced Differentiation Methods • Implicit differentiation • Logarithmic differentiation • Higher-order derivatives	Isolate derivatives in multi-variable coupled expressions, use log laws to simplify complex exponential expressions before differentiating.
L7	Elasticity of Demand & Root Finding • Price elasticity of demand (point elasticity) • Elasticity vs. Total Revenue relationships • Finding roots of equations using Newton's Method	Calculate demand elasticity to analyze pricing strategies, apply Newton's numerical iteration method to locate functional roots.
Unit 3: Curve Sketching & Applied Maxima/Minima (8 Hours)		
L8	Relative Extrema & Curve Analysis • Stationary and critical points • The First Derivative Test • Intervals of increase and decrease	Locate coordinates where a function's slope is zero or undefined, determine where curves rise or fall, isolate local turning points.
L9	Global Optimization & Concavity • Absolute extrema on a closed interval $[a, b]$ • Concavity (upward vs. downward) • Points of inflection (diminishing returns)	Locate definitive global turning points inside restricted operational boundaries, analyze direction shifts in curvature to locate points of diminishing returns.

L10	The Second Derivative Test & Asymptotes • Using second derivatives to classify extrema • Vertical, horizontal, and oblique asymptotes • Comprehensive functional curve sketching	Use second derivative signs at critical points to test for relative maximums or minimums, find structural boundaries of rational economic functions, draft exact visual plots.
L11	Applied Business Optimization • Formulating business word problems • Cost minimization and revenue maximization • Total profit optimization models	Translate narrative business challenges into algebraic optimization models, maximize corporate profit, minimize average production costs using derivatives.
Unit 4: Integration (9 Hours)		
L12	The Indefinite Integral & Initial Conditions • Integration as anti-differentiation • Finding the constant of integration (C) • Recovering total functions from marginal functions	Understand integration as the mathematical inverse of a derivative, solve for fixed constants given a benchmark baseline such as finding Total Cost from Marginal Cost.
L13	Advanced Formulas & Substitution Method • Integration formulas for exponential/reciprocal shapes • Techniques of integration: u-substitution • Summation notation ($\sum$) properties	Apply specialized integration formulas for non-algebraic expressions, simplify compound integrals by substituting variables for composite parts.
L14	The Definite Integral & Area • Bounded definite integration evaluation • The Fundamental Theorem of Integral Calculus • Calculating the area between intersecting curves	Evaluate definite integrations bounded by specified upper and lower numerical limits, link the concept of antiderivatives to physical area computations under curves.

L15	Economic Welfare Analysis • Defining market equilibrium parameters • Calculating Consumer's Surplus (CS) • Calculating Producer's Surplus (PS)	Determine precise mathematical areas between supply or demand equations and market prices, evaluate total societal welfare and market efficiency using integration.
Unit 5: Methods and Applications of Integration (6 Hours)		
L16	Advanced Integration Techniques • Integration by Parts • Integration by Partial Fractions	Apply the product rule in reverse to integrate equations with mismatched variables, deconstruct rational algebraic fractions into simpler, integrable components.
L17	Introduction to Differential Equations • Order and degree of ordinary differential equations • First-order, first-degree separable differential equations	Identify structures and target patterns of variable differential functions, isolate independent and dependent variables on opposite sides to find general solutions.
L18	Economic Applications of Differential Equations • Exponential growth and decay profiles • Newton's law of cooling • Related problems in economics	Construct and solve first-order differential equations representing commercial growth patterns and track asset depreciation curves over time.
Unit 6: Multivariable Calculus (10 Hours)		
L19	Functions of Several Variables & Partial Derivatives • Mapping 3D functional surfaces: $z = f(x, y)$ • First-order partial derivatives • Marginal productivity of Capital and Labor	Conceptualize and navigate equations with multiple independent input tracks, differentiate expressions with respect to one variable while holding others constant.

L20	Advanced Multi-Variable Calculus • Higher-order partial derivatives (cross-partials) • Checking cross-variable consistency (equality of mixed partials)	Compute secondary and cross-partial derivative layers, verify algebraic symmetry and cross-variable consistency in joint economic models.
L21	Unconstrained Optimization for Two Variables • Identifying critical points for multi-input systems • Extrema rules for two independent parameters	Locate multi-dimensional critical coordinates using joint first-order conditions, determine optimal independent variable values.
L22	The Second Derivative Test & Saddle Points • Constructing and solving the Hessian determinant • Classifying multi-variable maxima, minima, and saddle points	Apply second-order partial derivative conditions to identify local surfaces accurately, determine Hessian matrices.
L23	Constrained Optimization (Lagrange Multipliers) • Setting up the Lagrangian objective function • Maximizing utility/production under budget constraints	Optimize equations subject to real-world resource or financial restrictions, maximize utility, minimize production costs given constraints.
L24	Multiple Integrals & Course Review • Evaluating double integrals over rectangular regions • Overall course review and discussion	Compute areas and volumes under surfaces using successive integration layers.

2.3 Prescribed Learning Resources

Basic Text Book

- Haeussler, E. F., Jr., Paul, R. S., & Wood, R. (2005). *Introductory Mathematical Analysis for Business, Economics, and the Life and Social Sciences* (11th ed.). USA: Pearson Prentice Hall (p. 8).

Reference Books

- Singh, J.K. *Business Mathematics*. India: Himalaya Publishing House.
- Srinivasa, & George, D. *Business Mathematics and Statistics*. India: New Age International Publishers.

CHAPTER 3

EVALUATION SCHEME

The final grading reflects a continuous assessment paradigm distributed symmetrically between tracking regular engagement and a final formal examination:

3.1 In-Semester Evaluation

The In-Semester Evaluation constitutes 50% of the total grading weight for the Business Mathematics II course, serving as a dynamic, continuous assessment mechanism designed to track regular student engagement, practical execution, and conceptual mastery. Rather than relying solely on a high-stakes final examination, this framework balances theoretical mastery with hands-on applications, collaborative skill sets, and professional presentation capabilities.

Assessment Component	Weight	Focus Areas & Evaluation Criteria
Internal Ex-amination	20%	Mid-semester algebraic proficiency, formula retention and application.
Project Work	10%	Real-world financial or economic problem-solving.
Presentation	5%	Communication and technical explanation skills, peer interactions and question handling.
Assignments	10%	Periodic mathematical problem sheets, self-study commitment and deadline compliance
Attendance	5%	Regular physical classroom presence.
Total	50%	

3.2 End-Semester Evaluation

KUSOM exam unit will conduct this end-semester assessment. This will cover 50% of the total weight. 10% Objective questions and 40% subjective questions will be asked.

3.3 Grading System

Grade	A	A-	B+	B	B-	C+	C	C-	D	F
Grade Point	4	3.7	3.3	3.0	2.7	2.3	2.0	1.7	1.0	0
Marks (in%)	80-100	75-79	70-74	65-69	60-64	55-59	50-54	45-49	40-44	<40

CHAPTER 4

MODEL QUESTION



Kathmandu University
School of Management (KUSOM)
Final Assessment

Level: Bachelor

Program: BBA (2nd Sem)

Subject: Business Mathematics II

Year: 2026

F.M: 50

P.M: 23

Time: 2 hrs.

GROUP A: Multiple-Choice Questions

[10 × 1 Mark = 10 Marks]

1. Evaluate the following limit: $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$.
A) 0 B) 3 C) 6 D) Does not exist
2. Find the derivative of $y = 4x^3 - 5x^2 + 2x - 7$ with respect to x .
A) $12x^2 - 10x + 2$ B) $4x^2 - 5x + 2$ C) $12x^3 - 10x^2 + 2$ D) $12x^2 - 10x$
3. Differentiate the function $y = e^{3x}$ with respect to x .
A) e^{3x} B) $3e^{3x}$ C) $\frac{1}{3}e^{3x}$ D) $3e^x$
4. Find the critical points of the function $f(x) = x^2 - 8x + 12$.
A) $x = 2$ B) $x = 4$ C) $x = 6$ D) $x = 8$
5. Determine the vertical asymptote of the function $y = \frac{2x+1}{x-5}$.
A) $x = -1/2$ B) $x = 2$ C) $x = 5$ D) $y = 2$

6. Evaluate the indefinite integral: $\int (6x^2 - 2x) dx$.
 A) $2x^3 - x^2 + C$ B) $12x - 2 + C$ C) $3x^3 - 2x^2 + C$ D) $2x^3 - 2x^2 + C$
7. Compute the value of the definite integral: $\int_1^3 4x dx$.
 A) 8 B) 12 C) 16 D) 20
8. Find the general solution of the separable differential equation: $\frac{dy}{dx} = 5y$.
 A) $y = 5x + C$ B) $y = Ce^{5x}$ C) $y = e^{5x} + C$ D) $y = \frac{5}{2}x^2 + C$
9. Given $z = 3x^2y + 5xy^3$, find the first-order partial derivative $\frac{\partial z}{\partial x}$.
 A) $6xy + 5y^3$ B) $3x^2 + 15xy^2$ C) $6x + 15y^2$ D) $6xy + 15xy^2$
10. The total cost function for a production line is $C(x) = 5x^2 + 20x + 500$. Find the level of production x where the marginal cost (MC) is exactly equal to \$120.
 A) 5 B) 10 C) 12 D) 20

GROUP B: Subjective Analytical Questions [8 × 5 Marks = 40 Marks]

1. A firm uses a piecewise function to define its total revenue $R(x)$ (in thousands of dollars) based on selling x hundred items:

$$R(x) = \begin{cases} 2x^2 + 5 & \text{if } 0 \leq x < 2 \\ 4x + 1 & \text{if } x \geq 2 \end{cases}$$

Determine if the revenue function is continuous at the threshold point $x = 2$.

2. For the linear demand equation $p = 15 - q$, verify that demand is elastic when $p = 10$, is inelastic when $p = 5$, and has unit elasticity when $p = 7.5$.

OR

The total cost function of a factory is given by $C(q) = 0.1q^2 + 5q + 200$. Find the mathematical expression for the **Marginal Cost**. Evaluate the marginal cost when the production output is $q = 20$ units and interpret your result.

3. A manufacturing company finds that its price-demand relationship is given by the equation $p = 100 - 2q$. Find the total revenue function $R(q)$. Use this revenue function to determine the Marginal Revenue function and calculate its value when $q = 15$.

4. Sketch the graph of:

$$f(x) = 2x^3 - 6x^2 + 5$$

Apply the Second Derivative Test to clearly classify whether each critical point represents a relative maximum or a relative minimum.

5. A firm determines that its total profit function from producing and selling x units of a product is given by:

$$\Pi(x) = -0.5x^2 + 30x - 100$$

Find the exact level of production x that maximizes the firm's profit. Calculate the maximum profit value at this production level.

6. The demand function for a product is

$$p = f(q) = 100 - 0.05q$$

where p is the price per unit (in dollars) for q units. The supply function is

$$p = g(q) = 10 + 0.1q$$

Determine consumers' surplus and producers' surplus under market equilibrium.

7. In a certain town, the population at any time changes at a rate proportional to the population. If the population in 1990 was 60,000 and in 2000 was 64,000, find an equation for the population at time t , where t is the number of years past 1990. What is the expected population in 2010?
8. Suppose a firm has an order for 200 units of its product and wishes to distribute its manufacture between two of its plants, plant 1 and plant 2. Let q_1 and q_2 denote the outputs of plants 1 and 2, respectively, and suppose the total-cost function is given by

$$c = f(q_1, q_2) = 2q_1^2 + q_1q_2 + q_2^2 + 200$$

How should the output be distributed in order to minimize costs?